Student Project Task: Build a Mini Grammarly

AI-Powered Writing Assistant

Objective

Develop a simple AI-powered application inspired by Grammarly that can detect writing mistakes, suggest corrections, and predict the next word while the user types.

Problem Statement

Create a web-based AI Writing Assistant where a user can enter English text. The application should analyze the text and provide intelligent suggestions for improving it.

Example: `I has a car and I am goin to school.` → `I have a car and I am going to school.`

Next-word prediction: `I am going to` → **[school] [market] [university]**

Minimum Requirements

1	Text Input	Textbox/text editor for sentences or paragraphs.
2	Spelling Correction	Detect spelling mistakes and suggest corrections, e.g. goin → going.
3	Grammar Correction	Detect grammatical errors, e.g. I has a car → I have a car.
4	Next-Word Prediction	Predict likely next words, e.g. I am going to → school.
5	Suggestion Interface	Highlight errors, display suggestions, and allow corrections to be accepted.
6	AI/ML Component	Use an ML/DL/NLP model; do not build the complete system using only if-else rules.

Recommended Development Path

RNN → **LSTM** → **GRU** → **Transformer**. Students may start with a small pretrained model or train their own model using an appropriate text dataset. Explain the selected model and architecture in the report.

Deliverables

• Source code
• Dataset and/or pretrained model used
• Working application
• Short architecture report
• Screenshots
• 5–10 example sentences demonstrating performance
• Brief model comparison

Bonus Features

Multiple next-word suggestions • Auto-correction while typing • Sentence rewriting • Synonym suggestions • Vocabulary improvement • Readability score • Voice input • Urdu-English support • RNN vs LSTM vs GRU vs Transformer comparison

Suggested Project Title

AI Writing Assistant: A Mini Grammarly for Automatic Grammar Correction, Spell Checking, and Next-Word Prediction

Important

The objective is not to reproduce the commercial Grammarly system. Build a smaller educational prototype demonstrating how NLP and deep-learning techniques can create an intelligent writing assistant.